Depropanizer Column Ver 1 · OTS

DC3-V1-PE-LIN-001

Line list and line sizing

Rev 0 · Issued for teaching

Unit	Depropanizer Column Ver 1 · OTS
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Title	Line list and line sizing
Discipline	Process
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Revision record

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Note. *Numbers come from the simulator model (design case) and the assumptions listed; not for construction. Codes are named for the method only; no client, contractor, licensor or vendor is implied.*

Contents

1 · Purpose and basis 3
2 · Line list 4
3 · Worked example and calculation method 5
Engineering queries raised 6

1 · Purpose and basis

Line list of the C-201 unit: every line number of the P&ID DC3-V1-PID-001 rev D, plus three lines drawn without a number (proposed numbers, marked “prop.”). For each line the flow is taken from the HMB DC3-V1-PE-HMB-001 (or the model state for the reboiler loop and the overhead condensate), the density at the line temperature, and the velocity, ρv² and the friction gradient are checked against the sizing criterion of its duty in the three HMB cases. Design pressure and temperature are inherited from the item the line belongs to (design-basis rules, see DC3-V1-ME-EQL-001); the piping class follows from the ASME B16.5 rating at the design temperature.

Sizing criteria (assumed company practice)

Code	Duty	v min m/s	v max m/s	ΔP max kPa/100 m	ρv² max Pa
Ls	Liquid, pump suction	0.3	1.5	15	—
Ld	Liquid, pump discharge / pressure flow	0.6	3.0	45	—
Lg	Liquid, gravity drain (self-venting)	—	1.0	5	—
Lt	Liquid, thermosyphon reboiler inlet	0.3	1.5	5	—
V	Vapour, overhead / process	—	15.0	20	6,000
Vv	Vapour, vent gas	—	20.0	25	6,000
TP	Two-phase (feed, reboiler return)	2.0	15.0	30	10,000
S	LP steam	—	30.0	20	—
W	Cooling water	1.0	3.0	30	—
C	Steam condensate (before trap)	—	1.5	20	—

Piping classes and insulation codes

Class	Description
A1L	CS ASME class 150, LT (impact tested −46 °C), CA 3 mm — hydrocarbon (assumed)
B1L	CS ASME class 300, LT (impact tested −46 °C), CA 3 mm — hydrocarbon (assumed)
A1S	CS ASME class 150, CA 1.5 mm — LP steam (assumed)
A1W	CS ASME class 150, CA 3 mm — cooling water (assumed)
A1C	CS ASME class 150, CA 3 mm — steam condensate (assumed)
Insulation H	heat conservation, mineral wool 50 mm (assumed)
Insulation AC	anti-condensation 40 mm (cold service at −1 °C, assumed)
Insulation PP	personnel protection (surface > 60 °C within reach, assumed)
Insulation —	none

2 · Line list

Line	From → to	Service	Ph.	Crit.	Design case: kg/h · ρ kg/m³ · v m/s · ΔP kPa/100 m	OK	Max feed: v · ΔP	OK	Turndown v	Size needed (max feed)	Op. P MPa g / T °C	Design P / T / MDMT	Class	Ins.	Items on the line
6"-P-200	Battery limit → V-201	Feed from the upstream unit to V-201	L	Ls	31,809 · 548.5 · 0.86 · 2.1	✓	1.01 · 2.9	✓	0.58	6"	0.40 / −1	1.35 / 65 / −46	A1L	AC	—
6"-P-201	V-201 → P-201A/B suction	Feed pump suction header	L	Ls	31,809 · 548.5 · 0.86 · 2.1	✓	1.01 · 2.9	✓	0.58	6"	0.42 / −1	1.35 / 65 / −46	A1L	AC	MV-201/202
4"-P-202	P-201A/B → E-201 (FV-201, FE-201, XV-201)	Feed pump discharge to the preheater	L	Ld	31,809 · 548.5 · 1.96 · 17.4	✓	2.29 · 23.6	✓	1.31	4"	2.56 / −1	2.85 / 65 / −46	B1L	AC	FV-201, XV-201
6"-P-203	E-201 → C-201 tray 18	Preheated feed (two-phase)	2P	TP	31,809 · 62.4 · 7.60 · 18.7	✓	8.80 · 25.0	✓	5.08	6"	1.71 / 75	2.85 / 175 / −46	B1L	H	TIC-202
8"-P-204	C-201 top → E-202	Overhead vapour (PSV-201 branch)	V	V	40,798 · 41.3 · 8.51 · 10.5	✓	9.92 · 14.3	✓	5.67	8"	1.70 / 45	1.90 / 75 / −46	B1L	—	PT-210
6"-P-205	E-202 → V-202	Condensate to the reflux drum (gravity)	L	Lg	40,709 · 471.7 · 1.29 · 4.0	✗ v > 1	1.49 · 5.3	✗ v > 1, ΔP > 5	0.85	8"	1.69 / 42	1.90 / 75 / −46	B1L	—	TI-213
1"-P-206	V-202 → Fuel gas (PV-210, HV-210)	Non-condensable vent	V	Vv	89 · 26.0 · 1.72 · 3.5	✓	10.50 · 180.3	✗ ΔP > 25	0.67	1.5"	1.69 / 42	1.90 / 75 / −46	B1L	—	PV-210
6"-P-207	V-202 → P-202A/B suction	Reflux / product pump suction	L	Ls	40,709 · 471.7 · 1.29 · 4.0	✓	1.49 · 5.3	✓	0.85	6"	1.69 / 42	1.90 / 75 / −46	B1L	—	MV-211/212
4"-P-208	P-202A/B → Reflux tee / LV-203 → storage	Reflux / product pump discharge header	L	Ld	40,709 · 471.7 · 2.92 · 32.9	✓	3.37 · 43.6	✗ v > 3	1.92	6"	2.35 / 42	2.60 / 70 / −46	B1L	—	PDT-230
4"-P-209	Reflux tee (FV-204) → C-201 tray 27	Reflux	L	Ld	24,608 · 471.7 · 1.76 · 12.2	✓	2.07 · 16.6	✓	1.16	4"	2.35 / 42	2.60 / 70 / −46	B1L	—	FV-204, FE-204
4"-P-210	C-201 sump → Debutanizer (LV-202)	Bottoms on pressure to the debutanizer	L	Ld	15,619 · 459.7 · 1.15 · 5.2	✓	1.34 · 7.0	✓	0.77	3"	1.72 / 109	1.95 / 140 / −46	B1L	H	LV-202, FE-202
8"-P-211	C-201 sump → E-203 inlet	Thermosyphon reboiler feed	L	Lt	99,988 · 459.7 · 1.87 · 5.8	✗ v > 1.5, ΔP > 5	2.15 · 7.7	✗ v > 1.5, ΔP > 5	1.29	10"	1.72 / 109	1.95 / 140 / −46	B1L	H	—
10"-P-212	E-203 outlet → C-201 below tray 1	Thermosyphon reboiler return (two-phase)	2P	TP	99,988 · 129.7 · 4.21 · 6.5	✓	4.83 · 8.5	✓	2.91	8"	1.72 / 109	1.95 / 175 / −46	B1L	H	—
6"-LS-301	LP steam header → E-201 shell (TV-202, XV-202)	LP steam to the feed preheater	S	S	5,817 · 2.9 · 30.18 · 13.4	✗ v > 30	35.81 · 18.7	✗ v > 30	19.40	8"	0.45 / 156	0.70 / 175 / 0	A1S	H	TV-202, XV-202, HV-202
6"-LS-302	LP steam header → E-203 shell (TV-208, XV-209)	LP steam to the reboiler	S	S	3,790 · 2.9 · 19.66 · 5.8	✓	22.77 · 7.7	✓	13.38	6"	0.45 / 156	0.70 / 175 / 0	A1S	H	TV-208, XV-209, HV-209
10"-CW-401	CW supply → E-202 tubes	Cooling water supply	W	W	330,000 · 998.0 · 1.81 · 9.9	✓	1.81 · 9.9	✓	1.81	10"	0.45 / 15	0.70 / 75 / 0	A1W	—	HV-216, FE-216
10"-CW-402	E-202 tubes → CW return	Cooling water return	W	W	330,000 · 998.0 · 1.81 · 9.9	✓	1.81 · 9.9	✓	1.81	10"	0.45 / 27	0.70 / 75 / 0	A1W	—	TI-215
3"-P-213 prop.	LV-203 → Propane storage	Propane product rundown (not numbered on the P&ID — proposed)	L	Ld	16,100 · 471.7 · 1.99 · 21.6	✓	2.25 · 27.4	✓	1.31	3"	2.35 / 42	2.60 / 70 / −46	B1L	—	LV-203, FE-203, AI-219
2"-CD-501 prop.	E-201 shell → ST-201 → condensate	E-201 steam condensate (not numbered on the P&ID — proposed)	C	C	5,817 · 937.4 · 0.80 · 11.7	✓	0.95 · 16.4	✓	0.50	2"	0.26 / 140	0.70 / 175 / 0	A1C	PP	ST-201
2"-CD-502 prop.	E-203 shell → ST-202 → condensate	E-203 steam condensate (not numbered on the P&ID — proposed)	C	C	3,790 · 928.2 · 0.52 · 5.2	✓	0.61 · 6.8	✓	0.35	2"	0.30 / 144	0.70 / 175 / 0	A1C	PP	ST-202

Summary: Design case: 17 of 20 lines inside the criteria (outside: 6"-P-205, 8"-P-211, 6"-LS-301); Maximum feed: 15 of 20 lines inside the criteria (outside: 6"-P-205, 1"-P-206, 4"-P-208, 8"-P-211, 6"-LS-301); Turndown: 20 of 20 lines inside the criteria. Lines outside the criteria are raised as engineering queries (section 4); the P&ID sizes are preliminary (P&ID note 2).

3 · Worked example and calculation method

- Line 6"-P-203 (Preheated feed (two-phase)), design case: mass flow 31,809 kg/h, homogeneous two-phase density 62.4 kg/m³ (homogeneous two-phase, 67 mol % vapour).
- Q = 31,809 / 62.4 = 510.0 m³/h; Sch 40 ID 154.1 mm, A = 0.01865 m² → v = 7.60 m/s; ρv² = 3,598 Pa.
- Re = ρvD/μ = 730,055 (μ 0.1 mPa s); Darcy friction factor (Churchill, roughness 0.046 mm); ΔP/100 m = f·(100/D)·ρv²/2 = 18.66 kPa.
- Criterion TP (Two-phase (feed, reboiler return)): v 2–15 m/s, ΔP ≤ 30 kPa/100 m, ρv² ≤ 10000 Pa → inside the criteria.
- Liquid density: model molar volumes at 15 °C corrected −1.25 / −1.08 kg/m³ per K (C3 / C4); vapour density: real gas with Z 0.68–0.73; steam: saturated at the header pressure (Z 0.97); condensate: saturated water.
- Thermosyphon loop (8"-P-211, 10"-P-212): circulation = model boil-up ÷ 0.30 (assumed outlet quality). Overhead condensate and P-202 suction/discharge: the model condensation rate (reflux + product).

Engineering queries raised

Findings made while generating this document. They are open until answered by the discipline named; the Project & deliverables tab collects them for the whole unit.

No.	Subject	Query	Proposed answer	Status
EP-02	Line 6"-P-205 outside the sizing criteria	6"-P-205 (Condensate to the reflux drum (gravity)) fails the “Liquid, gravity drain (self-venting)” criteria in the design case / maximum feed: v > 1 (v 1.29 m/s, ΔP 3.99 kPa/100 m); v > 1, ΔP > 5 (v 1.49 m/s, ΔP 5.28 kPa/100 m).	Change to 8" (smallest size meeting the criteria in the maximum-feed case) or justify by a hydraulic check of the actual route; P&ID note 2 says the sizes are preliminary.	OPEN
EP-03	Line 1"-P-206 outside the sizing criteria	1"-P-206 (Non-condensable vent) fails the “Vapour, vent gas” criteria in the maximum feed: ΔP > 25 (v 10.5 m/s, ΔP 180.33 kPa/100 m).	Change to 1.5" (smallest size meeting the criteria in the maximum-feed case) or justify by a hydraulic check of the actual route; P&ID note 2 says the sizes are preliminary.	OPEN
EP-04	Line 4"-P-208 outside the sizing criteria	4"-P-208 (Reflux / product pump discharge header) fails the “Liquid, pump discharge / pressure flow” criteria in the maximum feed: v > 3 (v 3.37 m/s, ΔP 43.64 kPa/100 m).	Change to 6" (smallest size meeting the criteria in the maximum-feed case) or justify by a hydraulic check of the actual route; P&ID note 2 says the sizes are preliminary.	OPEN
EP-05	Line 8"-P-211 outside the sizing criteria	8"-P-211 (Thermosyphon reboiler feed) fails the “Liquid, thermosyphon reboiler inlet” criteria in the design case / maximum feed: v > 1.5, ΔP > 5 (v 1.87 m/s, ΔP 5.84 kPa/100 m); v > 1.5, ΔP > 5 (v 2.15 m/s, ΔP 7.68 kPa/100 m).	Change to 10" (smallest size meeting the criteria in the maximum-feed case) or justify by a hydraulic check of the actual route; P&ID note 2 says the sizes are preliminary.	OPEN
EP-06	Line 6"-LS-301 outside the sizing criteria	6"-LS-301 (LP steam to the feed preheater) fails the “LP steam” criteria in the design case / maximum feed: v > 30 (v 30.18 m/s, ΔP 13.38 kPa/100 m); v > 30 (v 35.81 m/s, ΔP 18.7 kPa/100 m).	Change to 8" (smallest size meeting the criteria in the maximum-feed case) or justify by a hydraulic check of the actual route; P&ID note 2 says the sizes are preliminary.	OPEN
EP-08	Product rundown and condensate lines not numbered	The propane rundown downstream of the reflux tee and LV-203, and the steam-condensate lines after ST-201 / ST-202, have no line numbers on DC3-V1-PID-001 rev D.	Number them 3"-P-213, 2"-CD-501 and 2"-CD-502 (sizes from the line list) in the next P&ID revision.	OPEN
EP-09	Thermosyphon circulation not in the model	The model computes the reboiler boil-up (520 kmol/h) but not the thermosyphon circulation; lines 8"-P-211 and 10"-P-212 were sized with 30 % mass vaporized at the outlet (assumed).	Confirm the circulation ratio with a thermosyphon rating (static head, outlet quality 15–35 %) before freezing the reboiler piping.	OPEN
EP-10	Volume basis of the flow meters	The model's liquid molar volumes are temperature independent (0.088 / 0.1034 m³/kmol); the HMB m³/h are therefore on a 15 °C basis while the hot bottoms (109 °C) occupy ≈ 22 % more volume.	State the reference temperature of every FI range and use actual density for meter sizing (instrument data sheets).	OPEN